



WORK SHEET - 31

QUADRATIC EQUATION & LOGARITHM

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.13 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

- Q.1 Let $f(x) = x^2 - 1$ for $0 < x < 2$ and $2x + 3$ for $2 \leq x < 3$. The quadratic equation whose roots are $\lim_{x \rightarrow 2^-} f(x)$ and $\lim_{x \rightarrow 2^+} f(x)$ is
(A) $x^2 - 4x + 21 = 0$ (B) $x^2 - 6x + 9 = 0$ (C) $x^2 - 10x + 21 = 0$ (D) $x^2 - 10x - 21 = 0$
- Q.2 Let $1, \omega$ and ω^2 be the cube roots of unity. The least possible degree of a polynomial, with real coefficients, having $2\omega^2, 3 + 4\omega, 3 + 4\omega^2$ and $5 - \omega - \omega^2$ as roots is
(A) 4 (B) 5 (C) 6 (D) 8
- Q.3 The roots of $(x - 41)^{49} + (x - 49)^{41} + (x - 2013)^{2013} = 0$ are
(A) all necessarily real.
(B) non-real except one positive real root.
(C) non-real except three positive real roots.
(D) non-real except for three real roots of which exactly one is positive.
- Q.4 Which of the following intervals is a possible domain of the function $f(x) = \log_{\{x\}} [x] + \log_{[x]} \{x\}$, where $[x]$ is the greatest integer not exceeding x and $\{x\}$ is the fractional part of x ?
(A) $(0, 1)$ (B) $(1, 2)$ (C) $(2, 3)$ (D) $(3, 5)$
- Q.5 Let a, b, c, d be numbers in the set $\{1, 2, 3, 4, 5, 6\}$ such that the curves $y = 2x^3 + ax + b$ and $y = 2x^3 + cx + d$ have no point in common. The maximum possible value of $(a - c)^2 + (b - d)$ is
(A) 3 (B) 4 (C) 5 (D) 6
- Q.6 Let n be a natural number and let a be a real number. The number of zeros of $x^{2n+1} - (2n+1)x + a = 0$ in the interval $[-1, 1]$ is
(A) 2 if $a > 0$ (B) 2 if $a < 0$
(C) at most one for every value of a (D) at least three for every value of a
- Q.7 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be the function $f(x) = (x - a_1)(x - a_2) + (x - a_2)(x - a_3) + (x - a_3)(x - a_1)$ with $a_1, a_2, a_3 \in \mathbb{R}$. Then $f(x) \geq 0 \forall x \in \mathbb{R}$ if and only if
(A) at least two of a_1, a_2, a_3 are equal (B) $a_1 = a_2 = a_3$
(C) a_1, a_2, a_3 are all distinct (D) a_1, a_2, a_3 are all positive and distinct
- Q.8 The set of values of ' p ' so that both the roots of the equation $(p - 5)x^2 - 2px + (p - 4) = 0$ are positive one is less than 2 and other is lying between 2 and 3, is
(A) $\left(\frac{49}{4}, 24\right)$ (B) $(5, \infty)$
(C) $(-\infty, 4) \cup \left(\frac{49}{4}, \infty\right)$ (D) $\left(5, \frac{49}{4}\right)$



- Q.9 If α, β are the roots of the equation $ax^2 + bx + c = 0$ then the roots of the equation $(a + b + c)x^2 - (b + 2c)x + c = 0$ are
- (A) $\frac{\alpha+1}{\alpha}, \frac{\beta+1}{\beta}$ (B) $\frac{\alpha-1}{\alpha}, \frac{\beta-1}{\beta}$ (C) $\frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1}$ (D) $\frac{\alpha}{\alpha-1}, \frac{\beta}{\beta-1}$
- Q.10 The ratio of the coefficient of x^{15} to the term independent of x in the expansion of $\left(x^2 + \frac{2}{x}\right)^{15}$ is
- (A) 1 : 4 (B) 1 : 32 (C) 7 : 64 (D) 7 : 16
- Q.11 Let A be a square matrix of order 3 such that $\det(A) = \frac{1}{2}$, then $\det(\text{adj } A^{-1})$ is equal to
- (A) 2 (B) 4 (C) 8 (D) 16
- Q.12 If α and β are real roots of the equation $x^2 + px + \frac{3p}{4} = 0$, such that $|\alpha - \beta| = \sqrt{10}$, then p belongs to the set
- (A) $\{-2, 5\}$ (B) $\{-3, 2\}$ (C) $\{3, -5\}$ (D) $\{2, -5\}$
- Q.13 The sum of the series $(2)^2 + 2(4)^2 + 3(6)^2 + \dots$ upto 10 terms is
- (A) 11300 (B) 12100 (C) 12300 (D) 11200

[PARAGRAPH TYPE]

Q.14 & Q.15 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Paragraph for question nos. 14 & 15

Consider the equations $(3x)^{\log 3} = (4y)^{\log 4}$ and $4^{\log x} = 3^{\log y}$

- Q.14 The value of x is
- (A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) 3 (D) 4
- Q.15 The value of y is
- (A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) 3 (D) 4

[MULTIPLE CORRECT CHOICE TYPE]

Q.16 & Q.17 has four choices (A), (B), (C), (D) out of which **ONE OR MORE** may be correct.

- Q.16 Let $f(x)$ be a quadratic expression with positive integral coefficients such that for every $\alpha, \beta \in \mathbb{R}$, $\beta > \alpha$, $\int_{\alpha}^{\beta} f(x) dx > 0$. Let $g(t) = f''(t) f(t)$ and $g(0) = 12$, then
- (A) 16 such quadratics of $g(x) = 0$ are possible (B) $f(x) = 0$ has non-real roots
(C) minimum value of $f(1)$ is 6 (D) maximum value of $f(1)$ is 11

Q.17 The equation $\log_{x^2} 16 + \log_{2x} 64 = 3$ has

- (A) one irrational solution
(C) two real solutions

- (B) no solution is a prime number
(D) one integral solution

[MATRIX TYPE]

Q.18 has four statements (A, B, C, D) given in Column-I and four statements (P, Q, R, S, T) given in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

Q.18	Column-I	Column-II
(A)	Let α and β be the roots of $2x^2 - ax + 4 = 0$ and $\sum_{n=1}^{\infty} \left(\frac{1}{\alpha} + \beta \right)^n = 3$, then 'a' is divisible by	(P) 3
(B)	Let $f(x) = 3x^2 + 7x \sin \alpha + \tan \beta$, where $\alpha, \beta \in (0, 2\pi)$, $f(2) = 15$. If $f(x) = 0$ has integral roots then number of ordered pairs of (α, β) is	(Q) 4
(C)	If area of the region formed by $y > \ln x$, $y < x$ and $0 < y < 1$ is $\left e - \frac{k}{2} \right $, then k is	(R) 5 (S) 8
(D)	With usual notations, in ΔABC , $r_1 = 3$, $r_2 = 6$ and $r_3 = 2$, then the value of 'a' is	(T) 9

[INTEGER TYPE]

Q.19 to Q.21 are "Integer Type" questions. (The answer to each of the questions are upto 4 digits)

Q.19 If a, b, c are distinct positive numbers in H.P. and if $\left(\frac{a+b}{2a-b} \right) + \left(\frac{c+b}{2c-b} \right) > \sqrt{\lambda} \sqrt{\lambda} \sqrt{\lambda} \dots \dots \dots \infty$, then find the greatest integral value of λ . Here $\lambda \geq 0$.

Q.20 Find the sum of solutions of equation $(5 + 2\sqrt{6})^{x^2-3} + (5 - 2\sqrt{6})^{x^2-3} = 10$.

Q.21 If $2x^{\log_4 3} + 3^{\log_4 x} = 27$, then $\sqrt{x}$ is equal to

ANSWER KEY

Q.1	C	Q.2	B	Q.3	B	Q.4	C	Q.5	C	Q.6	C	Q.7	B
Q.8	A	Q.9	D	Q.10	B	Q.11	B	Q.12	A	Q.13	B	Q.14	B
Q.15	A	Q.16	ABCD	Q.17	ABCD	Q.18	(A) P, T; (B) Q; (C) P; (D) Q						
Q.19	0004	Q.20	0000	Q.21	0004								